

Prüfbericht-Nr.: Test report no.:	CN2545AA 002	Auftrags-Nr.: Order no.:	180372789	Seite 1 von 34 Page 1 of 34
Kunden-Referenz-Nr.: Client reference no.:	N/A	Auftragsdatum: Order date:	2026.03.31	
Auftraggeber: Client:	Ningbo Longwin Technology Co., Ltd. No.7 Miaohoutou Road, Lanjiang Street, Yuyao City, 315400 Zhejiang P.R. China			
Prüfgegenstand: Test item:	Spot Cleaner			
Bezeichnung / Typ-Nr.: Identification / Type no.:	Refer to page 3.			
Auftrags-Inhalt: Order content:	TÜV Rheinland – EMC Service			
Prüfgrundlage: Test specification:	EN IEC 55014-1:2021 EN IEC 55014-2:2021 EN 61000-3-3:2013+A1+A2 EN IEC 61000-3-2:2019+A1+A2			
Wareneingangsdatum: Date of sample receipt:	2026.05.06			
Prüfmuster-Nr.: Test sample no.:	A004012758-001~002			
Prüfzeitraum: Testing period:	2026.05.13-2026.05.15			
Ort der Prüfung: Place of testing:	Refer to section 1.1			
Prüflaboratorium: Testing laboratory:	TÜV Rheinland / CCIC (Ningbo) Co., Ltd.			
Prüfergebnis*: Test result*:	Pass			
geprüft von: tested by:			genehmigt von: authorized by:	
Datum: Date:	2026.05.21		Ausstellungsdatum: Issue date:	2026.05.21
Stellung / Position:	Tracy Zhang/PE		Stellung / Position:	Carrie Lei/Authorizer
Sonstiges / Other:	Refer to page 3 for further information.			
Zustand des Prüfgegenstandes bei Anlieferung: Condition of the test item at delivery:	Prüfmuster vollständig und unbeschädigt Test item complete and undamaged			
* Legende:	P(ass) = entspricht o.g. Prüfgrundlage(n)	F(ail) = entspricht nicht o.g. Prüfgrundlage(n)	N/A = nicht anwendbar	N/T = nicht getestet
* Legend:	P(ass) = passed a.m. test specification(s)	F(ail) = failed a.m. test specification(s)	N/A = not applicable	N/T = not tested
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Anmerkungen
Remarks

1	Alle eingesetzten Prüfmittel waren zum angegebenen Prüfzeitraum gemäß eines festgelegten Kalibrierungsprogramms unseres Prüfhauses kalibriert. Sie entsprechen den in den Prüfprogrammen hinterlegten Anforderungen. Die Rückverfolgbarkeit der eingesetzten Prüfmittel ist durch die Einhaltung der Regelungen unseres Managementsystems gegeben. Detaillierte Informationen bezüglich Prüfkonditionen, Prüfequipment und Messunsicherheiten sind im Prüflabor vorhanden und können auf Wunsch bereitgestellt werden. <i>The equipment used during the specified testing period was calibrated according to our test laboratory calibration program. The equipment fulfils the requirements included in the relevant standards. The traceability of the test equipment used is ensured by compliance with the regulations of our management system.</i> <i>Detailed information regarding test conditions, equipment and measurement uncertainty is available in the test laboratory and could be provided on request.</i>
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3	Prüfklausel mit der Note * wurden an qualifizierte Unterauftragnehmer vergeben und sind unter der jeweiligen Prüfklausel des Berichts beschrieben. Abweichungen von Prüfspezifikation(en) oder Kundenanforderungen sind in der jeweiligen Prüfklausel im Bericht aufgeführt. <i>Test clauses with remark of * are subcontracted to qualified subcontractors and described under the respective test clause in the report. Deviations of testing specification(s) or customer requirements are listed in specific test clause in the report.</i>
4	Die Entscheidungsregel für Konformitätserklärungen basierend auf numerischen Messergebnissen in diesem Prüfbericht basiert auf der "Null-Grenzwert-Regel" und der "Einfachen Akzeptanz" gemäß ILAC G8:2019 und IEC Guide 115:2021, es sei denn, in der auf Seite 1 dieses Berichts genannten angewandten Norm ist etwas anderes festgelegt oder vom Kunden gewünscht. Dies bedeutet, dass die Messunsicherheit nicht berücksichtigt wird und daher auch nicht im Prüfbericht angegeben wird. Zu weiteren Informationen bezüglich des Risikos durch diese Entscheidungsregel siehe ILAC G8:2019. <i>The decision rule for statements of conformity, based on numerical measurement results, in this test report is based on the "Zero Guard Band Rule" and "Simple Acceptance" in accordance with ILAC G8:2019 and IEC Guide 115:2021, unless otherwise specified in the applied standard mentioned on Page 1 of this report or requested by the customer. This means that measurement uncertainty is not taken in account and hence also not declared in the test report. For additional information to the resulting risk based of this decision rule please refer to ILAC G8:2019.</i>

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-Model List:

No.	Model Name	Rated Voltage	Rated Power	Remark
1.	LW-F6001A	AC 220-240V, 50/60Hz	A1:450W, A2:550W, A3:800W	---
2.	LW-F6001C	AC 220-240V, 50/60Hz	C1:1650W, C2:1750W	---
3.	LW-F6001A1	AC 220-240V, 50/60Hz	450W	Have the same circuit diagram and PCB layout
4.	LW-F6002A1	AC 220-240V, 50/60Hz	450W	
5.	LW-F6001A2	AC 220-240V, 50/60Hz	550W	
6.	LW-F6002A2	AC 220-240V, 50/60Hz	550W	
7.	LW-F6001A3	AC 220-240V, 50/60Hz	800W	
8.	LW-F6002A3	AC 220-240V, 50/60Hz	800W	
9.	LW-F6001C1	AC 220-240V, 50/60Hz	1650W	Have the same circuit diagram and PCB layout
10.	LW-F6002C1	AC 220-240V, 50/60Hz	1650W	
11.	LW-F6001C2	AC 220-240V, 50/60Hz	1750W	
12.	LW-F6002C2	AC 220-240V, 50/60Hz	1750W	

Other aspects:

1. This report is to add new models and change the address of license holder.
2. Models LW-F6001A and LW-F6001C have been approved in test report CN2545AA 001.
3. As the client's requests, the address of license holder is changed as mentioned above.
4. In electrical characteristics, new models No. 3~8 are based on the same motor and motor control circuit. The differences among them are rated power marked on the label and mechanical aspect.
5. Models No. 9~12 have the same circuit diagram, PCB layout and motor. The differences among them are rated power marked on the label and mechanical aspect. They have hot water cleaning function.
6. Therefore, all EMC tests were performed as below table. The symbol "√" means the testing item was performed.

Model	DV	DP	Click	Har	Flick	ESD	EFT	Surge	CS	Dips
LW-F6002C2	√	√	√	√	√	√	√	√	√	√
LW-F6002A3	√	√		√	√	√	√	√	√	√

7. This test report is valid with test report CN2545AA 001.

TEST SUMMARY

5.1.1 HARMONICS ON AC MAINS

Result:

Pass

5.1.2 VOLTAGE CHANGES, VOLTAGE FLUCTUATIONS AND FLICKER ON AC MAINS

Result:

Pass

5.1.3 MAINS TERMINAL CONTINUOUS DISTURBANCE VOLTAGE

Result:

Pass

5.1.4 DISCONTINUOUS INTERFERENCE ON AC MAINS

Result:

Pass

5.2.1 DISTURBANCE POWER

Result:

Pass

5.2.2 RADIATED DISTURBANCE IN THE FREQUENCY RANGE FROM 30MHZ TO 1000MHZ

Result:

Pass

6.1.1 ELECTROSTATIC DISCHARGE

Result:

Pass

6.2.1 FAST TRANSIENTS ON AC POWER LINES

Result:

Pass

6.2.2 INJECTED CURRENT INTO AC POWER PORT

Result:

Pass

6.2.3 SURGES TO AC POWER PORT

Result:

Pass

6.2.4 VOLTAGE DIPS AND INTERRUPTIONS TO AC POWER PORT

Result:

Pass

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1 Test Sites

1.1 Test Facilities

Laboratory: TÜV Rheinland /CCIC(Ningbo) Co., Ltd.

1st Floor, Building 11, Scholar Innovation Park, No.1188 Zhongguan Road, Zhenhai District, Ningbo 315200 P.R. China.

The used test equipment is in accordance with CISPR 16-1 series standards for measurement of radio interference.

The tests performed in Laboratory were under supervision of TÜV Rheinland/CCIC's engineer.

1.2 List of Test and Measurement Instruments

Table 1: List of Test and Measurement Equipment of Laboratory

No.	Equipment	Model	Serial no.	Cal. due date
1.	EMI test receiver	ESR3	102331	2026.10.22
2.	LISN	ENV216	102250	2026.10.22
3.	Click Analyzer	DDA55	14041726085	2026.10.21
4.	LISN	LS16C/10	16011726408	2026.10.21
5.	EMI test receiver	ESR3	102362	2026.10.22
6.	Absorbing clamp	MDS-21	100857	2026.10.28
7.	Harmonic power supply	5001IX-CTS-400-413	1735A02292	2026.11.17
8.	Harmonics/voltage fluctuation Analyser	PACS-1	1736A00799	2026.11.17
9.	ESD generator	NX30.1	11744	2026.11.19
10.	CS immunity test	NSG 4070C-80	47606	2026.10.22
11.	CDN	CDN M016S	47812	2026.10.24
12.	Attenuator	JTH-SJ-100W-6dB	100145143000 573	N/A
13.	EFT/Surge/PFM/Dips Immunity test system	Compact NX5 bsp-1-300-16	P1725200261	2026.10.21

1.3 Measurement Uncertainty

Table 2: Measurement Uncertainty of Laboratory

Test Item	Expanded Measurement Uncertainty (k=2)
Conducted Emission (9-150kHz)	3.70dB
Conducted Emission (150k-30MHz)	3.30dB
Disturbance Power	4.27dB
Radiated Emission (30-1000MHz)	4.39dB
Radiated Emission (1-18GHz)	4.67dB
Radiated Emission (CDNE method)	4.05dB

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2 General Product Information

2.1 Product Function and Intended Use

The EUT (equipment under test) is an ordinary Spot Cleaner for household and similar use. For the further information, refer to the user's manual.

2.2 Ratings and System Details

System input voltage	:	Refer to page 3.	For all models
Rated power	:	Refer to page 3.	For all models
Protection Class	:	Class I	for model series with C
		Class II	for model series with A

Refer to the User's Manual for further information.

2.3 Independent Operation Modes

The basic operation modes are: "On" or "Off", model series with C have cold water cleaning and hot water cleaning. Model series with A only have cold water cleaning.

Refer to the user's manual for further information.

2.4 Noise Generating and Noise Suppressing Parts

Refer to the circuit diagram for further information.

2.5 Submitted Documents

Circuit diagram, PCB, label and user's manual etc.

3 Test Set-up and Operation Modes

3.1 Principle of Configuration Selection

Emission: The equipment under test (EUT) was configured to measure its highest possible radiation level. The test conditions were adapted accordingly in reference to the instructions for use.

Refer to the related paragraph of this report.

Immunity: The equipment under test (EUT) was configured to have its highest possible susceptibility against the tested phenomena. The test conditions were adapted accordingly in reference to the instructions for use.

Refer to the related paragraph of this report.

3.2 Physical Configuration for Testing

Refer to the related paragraph of this report.

3.3 Test Operation and Test Software

Refer to the related paragraph of this report. No software was used.

3.4 Special Accessories and Auxiliary Equipment

None.

3.5 Countermeasures to achieve EMC Compliance

The tested sample contained noise suppression components as described in the circuit diagram. No special measure is employed to achieve the requirement.

4 Conformity Decision Rule

For all EMI tests (when included in this report), as measurement uncertainties are less than the values U_{CISPR} given in CISPR 16-4-2, compliance with the limits is determined by comparing measurement results directly with corresponding limits without taking into consideration of measurement uncertainties. For all EMS tests (when included in this report), measurement uncertainties are not considered as well according to corresponding test standards.

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5 Test Results EMISSION

5.1 Emission in the Frequency Range up to 30 MHz

5.1.1 Harmonics on AC Mains

Result:	Pass
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Date of testing : 2026.05.13-2026.05.14
Test procedure : EN IEC 61000-3-2:2019+A1+A2
Test duration : 2.5min
Harmonic order : 2 – 40th
Frequency range : 0 – 2kHz
Test voltage : AC 230V, 50Hz
Ambient Condition : Temperature: 20 °C; Relative Humidity: 57%

The harmonics on AC Mains in the frequency from 0 to 2 kHz were measured in accordance with EN IEC 61000-3-2:2019+A1+A2.

The measurement was conducted with an automatic current harmonic analyzing system. This equipment is in compliance with the requirements of EN IEC 61000-3-2:2019+A1+A2.

The results indicated in the following tables and figures were those measured and recorded by an automatic measuring system.

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Table 3: Harmonic currents measurement result for model LW-F6002C2

Equipment category: Class A;

Fundamental current I₁: 6.297A; Power factor: 0.990; Active input power: 1440.9W.

Harm#	Harms(avg)	100%Limit	%of Limit	Harms(max)	150%Limit	%of Limit	Status
2	0.376	1.080	34.9	0.416	1.620	25.7	Pass
3	0.637	2.300	27.7	0.689	3.450	20.0	Pass
4	0.306	0.430	71.1	0.335	0.645	52.0	Pass
5	0.158	1.140	13.9	0.174	1.710	10.2	Pass
6	0.216	0.300	72.1	0.234	0.450	51.9	Pass
7	0.009	0.770	N/A	0.018	1.155	N/A	Pass
8	0.129	0.230	56.3	0.138	0.345	39.9	Pass
9	0.069	0.400	17.1	0.075	0.600	12.5	Pass
10	0.062	0.184	33.8	0.068	0.276	24.5	Pass
11	0.056	0.330	16.9	0.060	0.495	12.1	Pass
12	0.020	0.153	N/A	0.024	0.230	N/A	Pass
13	0.055	0.210	26.1	0.057	0.315	18.1	Pass
14	0.007	0.131	N/A	0.008	0.197	N/A	Pass
15	0.030	0.150	N/A	0.034	0.225	N/A	Pass
16	0.014	0.115	N/A	0.015	0.173	N/A	Pass
17	0.021	0.132	N/A	0.025	0.198	N/A	Pass
18	0.016	0.102	N/A	0.018	0.153	N/A	Pass
19	0.008	0.118	N/A	0.011	0.178	N/A	Pass
20	0.015	0.092	N/A	0.018	0.138	N/A	Pass
21	0.005	0.107	N/A	0.007	0.161	N/A	Pass
22	0.011	0.084	N/A	0.013	0.125	N/A	Pass
23	0.006	0.098	N/A	0.008	0.147	N/A	Pass
24	0.005	0.077	N/A	0.008	0.115	N/A	Pass
25	0.008	0.090	N/A	0.011	0.135	N/A	Pass
26	0.003	0.071	N/A	0.005	0.107	N/A	Pass
27	0.008	0.083	N/A	0.012	0.125	N/A	Pass
28	0.007	0.066	N/A	0.010	0.099	N/A	Pass
29	0.005	0.078	N/A	0.010	0.116	N/A	Pass
30	0.011	0.061	N/A	0.012	0.092	N/A	Pass
31	0.003	0.073	N/A	0.005	0.109	N/A	Pass
32	0.011	0.058	N/A	0.013	0.086	N/A	Pass
33	0.004	0.068	N/A	0.006	0.102	N/A	Pass
34	0.010	0.054	N/A	0.012	0.081	N/A	Pass
35	0.008	0.064	N/A	0.010	0.096	N/A	Pass
36	0.007	0.051	N/A	0.012	0.077	N/A	Pass
37	0.008	0.061	N/A	0.012	0.091	N/A	Pass
38	0.004	0.048	N/A	0.011	0.073	N/A	Pass
39	0.009	0.058	N/A	0.012	0.087	N/A	Pass
40	0.002	0.046	N/A	0.009	0.069	N/A	Pass

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Table 4: Harmonic currents measurement result for model LW-F6002A3

Equipment category: Class A;

Fundamental current I₁: 2.356A; Power factor: 0.982; Active input power: 540.0W.

Harm#	Harms(avg)	100%Limit	%of Limit	Harms(max)	150%Limit	%of Limit	Status
2	0.002	1.080	N/A	0.003	1.620	N/A	Pass
3	0.421	2.300	18.3	0.488	3.450	14.1	Pass
4	0.001	0.430	N/A	0.002	0.645	N/A	Pass
5	0.061	1.140	5.4	0.067	1.710	3.9	Pass
6	0.001	0.300	N/A	0.001	0.450	N/A	Pass
7	0.018	0.770	2.3	0.022	1.155	1.9	Pass
8	0.001	0.230	N/A	0.003	0.345	N/A	Pass
9	0.013	0.400	N/A	0.017	0.600	N/A	Pass
10	0.002	0.184	N/A	0.013	0.276	N/A	Pass
11	0.011	0.330	N/A	0.020	0.495	N/A	Pass
12	0.002	0.153	N/A	0.014	0.230	N/A	Pass
13	0.010	0.210	N/A	0.021	0.315	N/A	Pass
14	0.001	0.131	N/A	0.008	0.197	N/A	Pass
15	0.010	0.150	N/A	0.013	0.225	N/A	Pass
16	0.001	0.115	N/A	0.003	0.173	N/A	Pass
17	0.007	0.132	N/A	0.009	0.198	N/A	Pass
18	0.001	0.102	N/A	0.004	0.153	N/A	Pass
19	0.007	0.118	N/A	0.009	0.178	N/A	Pass
20	0.001	0.092	N/A	0.006	0.138	N/A	Pass
21	0.006	0.107	N/A	0.010	0.161	N/A	Pass
22	0.003	0.084	N/A	0.028	0.125	N/A	Pass
23	0.005	0.098	N/A	0.010	0.147	N/A	Pass
24	0.003	0.077	N/A	0.030	0.115	N/A	Pass
25	0.004	0.090	N/A	0.005	0.135	N/A	Pass
26	0.001	0.071	N/A	0.008	0.107	N/A	Pass
27	0.003	0.083	N/A	0.004	0.125	N/A	Pass
28	0.001	0.066	N/A	0.003	0.099	N/A	Pass
29	0.002	0.078	N/A	0.005	0.116	N/A	Pass
30	0.001	0.061	N/A	0.001	0.092	N/A	Pass
31	0.002	0.073	N/A	0.002	0.109	N/A	Pass
32	0.001	0.058	N/A	0.002	0.086	N/A	Pass
33	0.001	0.068	N/A	0.002	0.102	N/A	Pass
34	0.001	0.054	N/A	0.002	0.081	N/A	Pass
35	0.001	0.064	N/A	0.004	0.096	N/A	Pass
36	0.000	0.051	N/A	0.001	0.077	N/A	Pass
37	0.001	0.061	N/A	0.003	0.091	N/A	Pass
38	0.000	0.048	N/A	0.001	0.073	N/A	Pass
39	0.001	0.058	N/A	0.001	0.087	N/A	Pass
40	0.000	0.046	N/A	0.001	0.069	N/A	Pass

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5.1.2 Voltage changes, voltage fluctuations and flicker on AC mains

Result:	Pass
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Date of testing : 2026.05.13-2026.05.14
 Test procedure : EN 61000-3-3:2013+A1+A2
 Ambient Condition : Temperature: 20 °C; Relative Humidity: 57%

According to the characteristics of the sample, as specified by clause 5 of the basic standard, following limits apply:

- the value of P_{st} shall not be greater than 1.0;
- the value of $d(t)$ during a voltage change shall not exceed 3.3% for more than 500ms;
- the relative steady-state voltage change, d_c , shall not exceed 3.3%;
- the maximum relative voltage change d_{max} , shall not exceed 6%.

Following are the measurement results obtained via an automatic testing system.

According to Annex A.7 of the standard EN 61000-3-3:2013+A1+A2, vacuum cleaner do not need to be evaluated for Plt.

Table 5: Voltage fluctuations and flicker measurement results for model LW-F6002C2

	d_c	$d_{max} (average)$	$d(t)$	P_{st}	P_{lt}
Limits	3.3%	6%	3.3%/500ms	1.0	N/A
Result	1.25%	1.92%	0ms	0.481	N/A

Table 6: Voltage fluctuations and flicker measurement results for model LW-F6002A3

	d_c	$d_{max} (average)$	$d(t)$	P_{st}	P_{lt}
Limits	3.3%	6%	3.3%/500ms	1.0	N/A
Result	0.44%	2.40%	0ms	0.586	N/A

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5.1.3 Mains Terminal Continuous Disturbance Voltage

Result:	Pass
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Date of testing : 2026.05.13
Kind of test site : EMC Shielding Room
Port : Mains
Basic Standard : EN IEC 55014-1:2021
Frequency Range : 150kHz – 30MHz
Limit : EN IEC 55014-1:2021, Clause 4.3.3 Table5
Ambient Condition : Temperature: 21 °C; Relative Humidity: 63%

Test Setup

Input Voltage : AC 230V, 50Hz
Operational mode : On
Artificial hand : Yes for LW-F6002A3
Earthing : No for LW-F6002A3; Yes for LW-F6002C2
Test Setup : According to Clause 5 of EN IEC 55014-1:2021

The measurement setup was made according to EN IEC 55014-1:2021 in an EMC shielding room.

The measurement equipment like test receiver, quasi-peak detector and Artificial Mains Network (AMN) are in compliance with CISPR 16-1 series standards and EN IEC 55014-1:2021. The tested object was operated under its rated voltage and its rated frequency. Prior to the measurements the test object operated about 15 minutes (warm-up) in order to stabilize its operating conditions and to ensure reliable measurement values.

Furthermore an internal calibration with the test receiver was conducted prior to and after each measurement.

The tested object was set-up on a wooden table and 0.8m away from the AMN. The length of power cord of EUT plus that of the extension cord was 1.5m. The EUT was set 0.8m away from the AMN. The cord longer than necessary to be connected to the AMN was folded forth and back parallel so as to form a bundle with a length between 0.3m and 0.4m.

The Disturbance Voltage was determined according to according to clause 5 of EN IEC 55014-1:2021 while measuring the anode and cathode conductor by turns.

The following figures were those measured by an automatic measuring system. The disturbance voltage was scanned firstly with both Peak and Average detector and then a final measurement was performed with both Quasi-peak and Average detector at the frequencies which showed the Max. in a designated frequency sub-range. In the figures below, the higher curve is that of peak-value and the lower one is average-value. The symbol “◆” refers to Quasi-peak value and the Average value which were measured in the final measurement.

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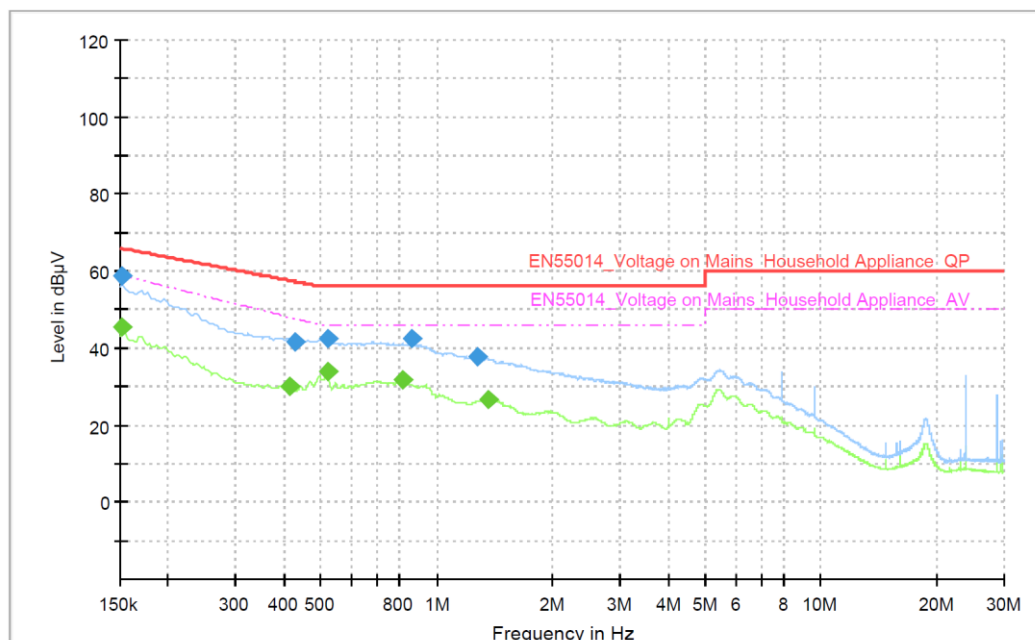
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Figure 1: Spectral Diagrams, Conducted Emission, 150kHz - 30MHz, L for LW-F6002C2

Full Spectrum



Final Result

Frequency (MHz)	QuasiPeak (dBµV)	CAverage (dBµV)	Limit (dBµV)	Margin (dB)	Meas. Time (ms)	Bandwidth (kHz)	Line	Filter	Corr. (dB)
0.152250	---	45.30	58.84	13.54	1000.0	9.000	L1	ON	9.9
0.152250	58.66	---	65.88	7.21	1000.0	9.000	L1	ON	9.9
0.415500	---	30.25	48.00	17.75	1000.0	9.000	L1	ON	9.9
0.426750	41.80	---	57.32	15.52	1000.0	9.000	L1	ON	9.9
0.519000	42.49	---	56.00	13.51	1000.0	9.000	L1	ON	9.9
0.519000	---	34.16	46.00	11.84	1000.0	9.000	L1	ON	9.9
0.813750	---	31.73	46.00	14.27	1000.0	9.000	L1	ON	9.9
0.858750	42.65	---	56.00	13.35	1000.0	9.000	L1	ON	9.9
1.279500	37.60	---	56.00	18.40	1000.0	9.000	L1	ON	9.9
1.356000	---	26.84	46.00	19.16	1000.0	9.000	L1	ON	9.9

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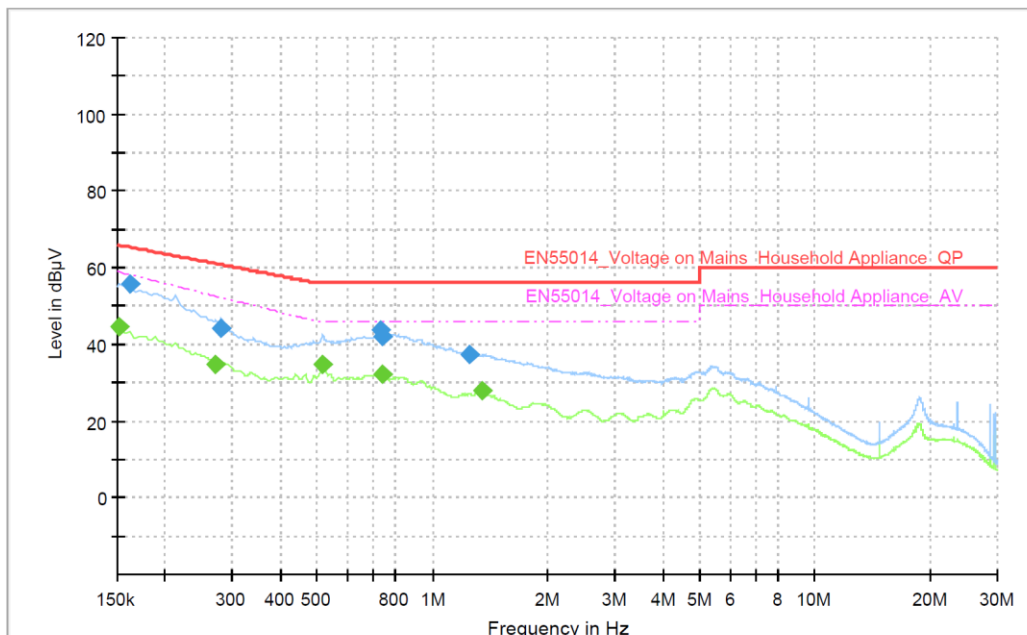
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Figure 2: Spectral Diagrams, Conducted Emission, 150kHz - 30MHz, N for LW-F6002C2

Full Spectrum



Final Result

Frequency (MHz)	QuasiPeak (dBµV)	CAverage (dBµV)	Limit (dBµV)	Margin (dB)	Meas. Time (ms)	Bandwidth (kHz)	Line	Filter	Corr. (dB)
0.152250	---	44.67	58.84	14.17	1000.0	9.000	N	ON	9.7
0.161250	55.61	---	65.40	9.79	1000.0	9.000	N	ON	9.7
0.271500	---	34.78	52.59	17.82	1000.0	9.000	N	ON	9.7
0.280500	44.09	---	60.80	16.71	1000.0	9.000	N	ON	9.7
0.516750	---	34.79	46.00	11.21	1000.0	9.000	N	ON	9.7
0.735000	43.64	---	56.00	12.36	1000.0	9.000	N	ON	9.7
0.737250	41.96	---	56.00	14.04	1000.0	9.000	N	ON	9.7
0.737250	---	32.05	46.00	13.95	1000.0	9.000	N	ON	9.7
1.250250	37.32	---	56.00	18.68	1000.0	9.000	N	ON	9.8
1.344750	---	27.83	46.00	18.17	1000.0	9.000	N	ON	9.8

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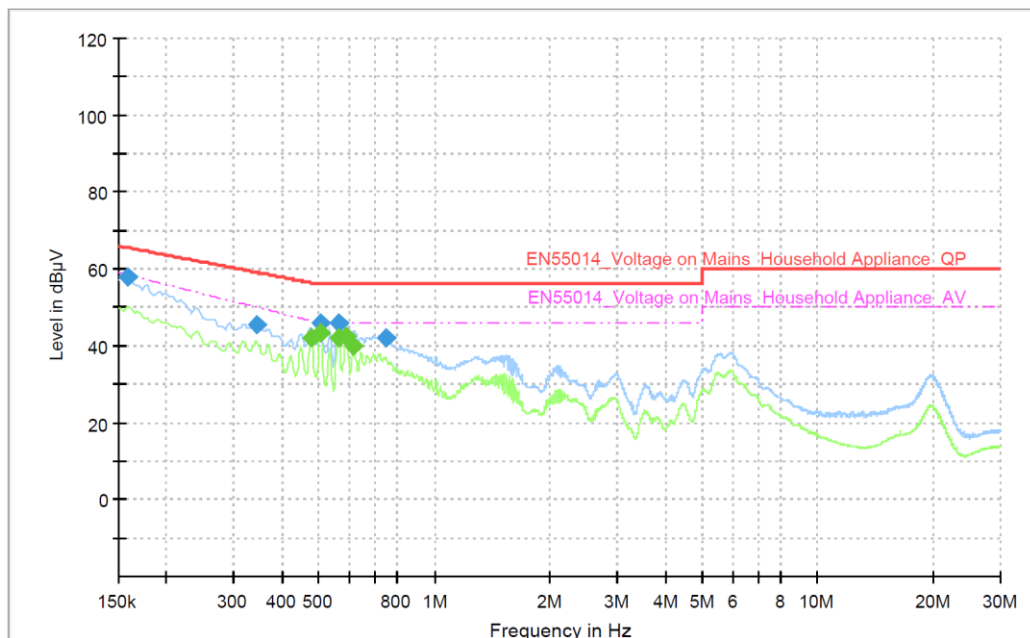
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Figure 3: Spectral Diagrams, Conducted Emission, 150kHz - 30MHz, L for LW-F6002A3

Full Spectrum



Final Result

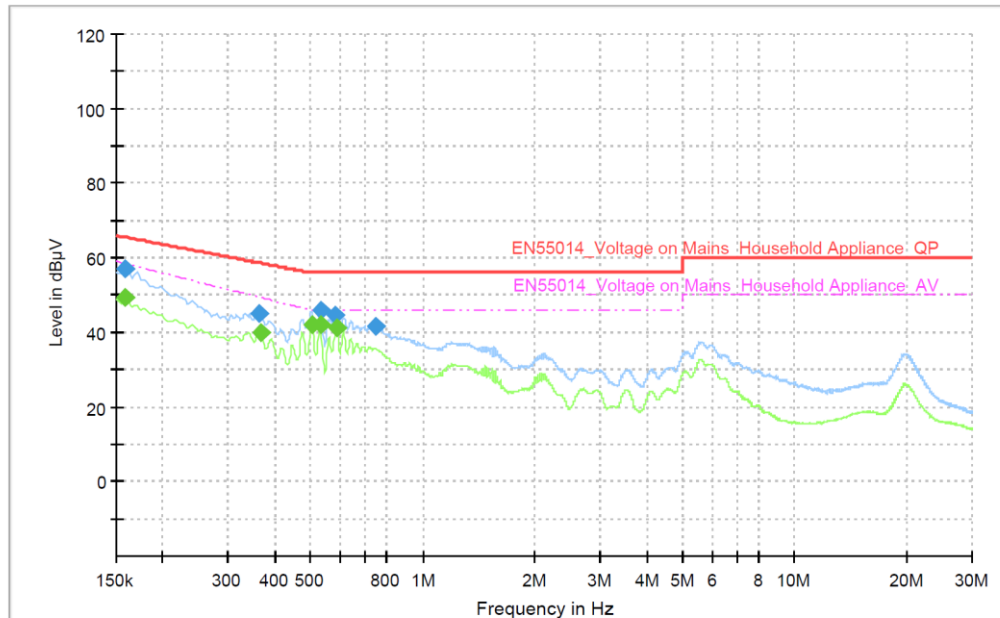
Frequency (MHz)	QuasiPeak (dBμV)	CAverage (dBμV)	Limit (dBμV)	Margin (dB)	Meas. Time (ms)	Bandwidth (kHz)	Line	Filter	Corr. (dB)
0.159000	57.97	---	65.52	7.55	1000.0	9.000	L1	ON	9.9
0.345750	45.41	---	59.06	13.66	1000.0	9.000	L1	ON	9.9
0.476250	---	42.00	46.53	4.52	1000.0	9.000	L1	ON	9.9
0.503250	---	43.40	46.00	2.60	1000.0	9.000	L1	ON	9.9
0.503250	45.86	---	56.00	10.14	1000.0	9.000	L1	ON	9.9
0.559500	45.86	---	56.00	10.14	1000.0	9.000	L1	ON	9.9
0.561750	---	42.09	46.00	3.91	1000.0	9.000	L1	ON	9.9
0.588750	---	42.52	46.00	3.48	1000.0	9.000	L1	ON	9.9
0.615750	---	40.14	46.00	5.86	1000.0	9.000	L1	ON	9.9
0.744000	42.02	---	56.00	13.98	1000.0	9.000	L1	ON	9.9

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Figure 4: Spectral Diagrams, Conducted Emission, 150kHz - 30MHz, N for LW-F6002A3

Full Spectrum



Final Result

Frequency (MHz)	QuasiPeak (dBμV)	CAverage (dBμV)	Limit (dBμV)	Margin (dB)	Meas. Time (ms)	Bandwidth (kHz)	Line	Filter	Corr. (dB)
0.159000	---	49.38	58.37	8.99	1000.0	9.000	N	ON	9.7
0.159000	57.13	---	65.52	8.38	1000.0	9.000	N	ON	9.7
0.361500	45.06	---	58.69	13.64	1000.0	9.000	N	ON	9.7
0.366000	---	40.15	49.37	9.22	1000.0	9.000	N	ON	9.7
0.503250	---	42.08	46.00	3.92	1000.0	9.000	N	ON	9.7
0.530250	45.84	---	56.00	10.16	1000.0	9.000	N	ON	9.7
0.530250	---	41.95	46.00	4.05	1000.0	9.000	N	ON	9.7
0.584250	44.76	---	56.00	11.24	1000.0	9.000	N	ON	9.7
0.586500	---	41.17	46.00	4.83	1000.0	9.000	N	ON	9.7
0.750750	41.51	---	56.00	14.49	1000.0	9.000	N	ON	9.7

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5.1.4 Discontinuous Interference on AC Mains

Result:	Pass
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Date of testing : 2026.05.13
 Test procedure : EN IEC 55014-1:2021 and CISPR 16-1 series standards
 Kind of test site : EMC shielded room
 Port : Mains
 Basic standard : EN IEC 55014-1:2021
 Frequency range : 0.15 – 30MHz
 Limit : EN IEC 55014-1:2021, clause 4.4
 Operating Condition : EN IEC 55014-1:2021, clause A.4
 Ambient Condition : Temperature: 23 °C ; Relative Humidity: 55 %

Measuring configuration and description

The discontinuous interference on AC mains in the frequency range from 0.15 to 30MHz was measured in accordance to EN IEC 55014-1:2021.

The measurement setup was made according to EN IEC 55014-1:2021. The used measurement equipment was in accordance to CISPR 16-1 series standards.

The test setup is according to clause A.4 of EN IEC 55014-1:2021.

The clicks were measured when the thermostat of the EUT started or stopped.

The clicks were measured at the frequency of 0.15MHz, 0.5MHz, 1.4MHz and 30MHz.

The tests include RUN A and RUN B. The first one is to detect the Click rate and RUN B is to detect how many clicks overtop the limits that are calculated according the formula below.

For $0.2 \leq N < 30$ the Sensitivity = RUN A + $20 \cdot \log(30/\text{Click rate})$ and for $N < 0.2$ the Sensitivity = RUN A + 44.

Table 7: Click Test Results of RUN A, for LW-F6002C2

Measured Frequency (MHz)	0.15	0.5	1.4	30
Sensitivity(dBuV)	66.0	56.0	56.0	60.0
Last Time T(min./sec.)	120/00	120/00	120/00	120/00
Click Number n1 of <20ms	25	6	0	0
Click Number n2 of >20ms	0	0	0	0
Total Click Number n=n1+n2	25	6	0	0
Click Rated $N=F \times n/T$	0.21	0.05		

According to the clause 5.4.3.4 of EN IEC 55014-1:2021:

- “
- the click rate is not more than 5,
 - none of the caused clicks has a duration longer than 20ms,
 - 90% of the caused clicks have a duration less than or equal 10ms

shall be verified at one frequency only, 150kHz or 500kHz, at which the higher click rate occurs. If any of these conditions is not satisfied then general assessment in accordance with 5.4.2 applies.”

Therefore, all models are deemed to fulfill the relevant requirements.

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5.2 Emission in the Frequency Range above 30 MHz

5.2.1 Disturbance Power

Result:	Pass
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Date of testing : 2026.05.14
Port : Mains
Basic Standard : EN IEC 55014-1:2021
Frequency Range : 30 – 300MHz
Limit : EN IEC 55014-1:2021, clause 4.3.4.4, Household appliance
Ambient : Temperature: 20°C; Relative Humidity: 45%
Condition

Test Setup

Input Voltage : AC 230V, 50Hz
Operational mode : ON
Earthing : No for LW-F6002A3; Yes for LW-F6002C2

Measuring configuration and description

The measurement setup was made according to EN IEC 55014-1:2021.

The measurement equipment like test receiver and absorption clamp are in compliance with CISPR 16-1 series standards. The test object has been operated by its rated voltage. Prior to the measurements the test objects operated about 10 minutes (warm-up) in order to stabilize their operating conditions and to ensure reliable measurement values.

Furthermore an internal calibration with the test receiver was conducted prior to each measurement.

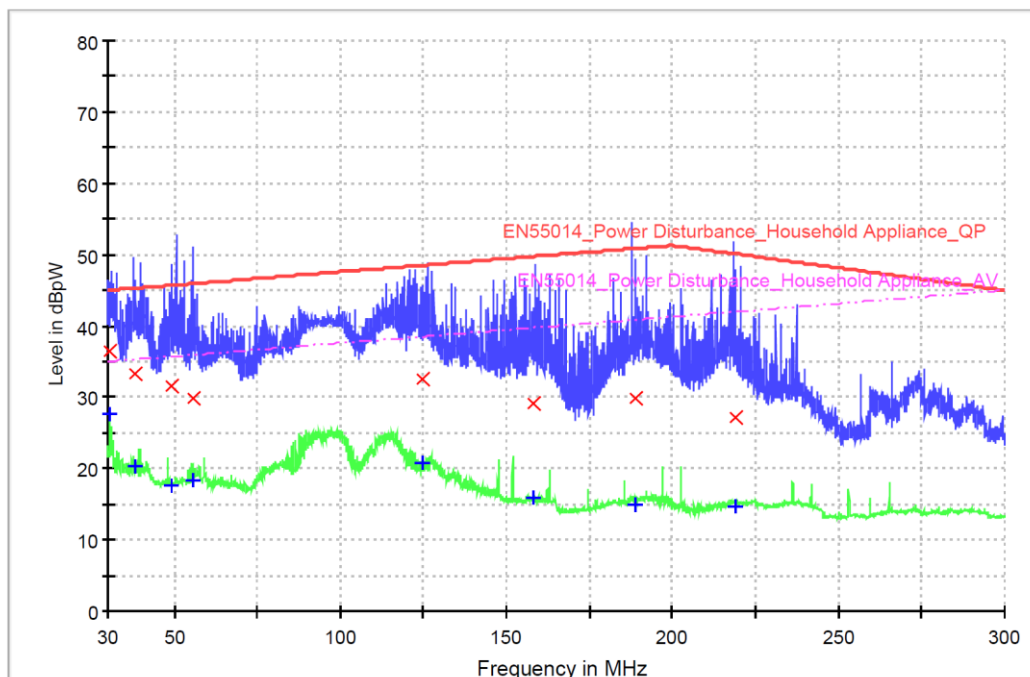
Before measurement, a survey was made to determine in which state the maximum disturbance was obtained. And the measurement was made in the state the maximum disturbance was obtained.

The tested object was set-up on a wooden bench.

The Disturbance Power was determined according to clause 5 of EN IEC 55014-1:2021. The length of power cord of EUT plus that of the extension cord was 6.0m.

The measurement was performed by operating the EUT in normal operation mode. The figures and tables below were those measured in the operation mode. Both Quasi Peak and Average Value were measured. In final measurement, by moving the absorption clamp along the power supply cord and the extension-power cord from the test object, Quasi-Peak and Average Value were measured and listed respectively where they had a maximum in previous scanning survey. In the Figures, the symbol “x” with red colour means Quasi-Peak Value and the symbol “+” with blue colour means Average Value which was measured in final measurement.

Figure 5: Spectral Diagrams, Power Disturbance, AC Mains, 30–300MHz for LW-F6002C2



Limit and Margin-QP

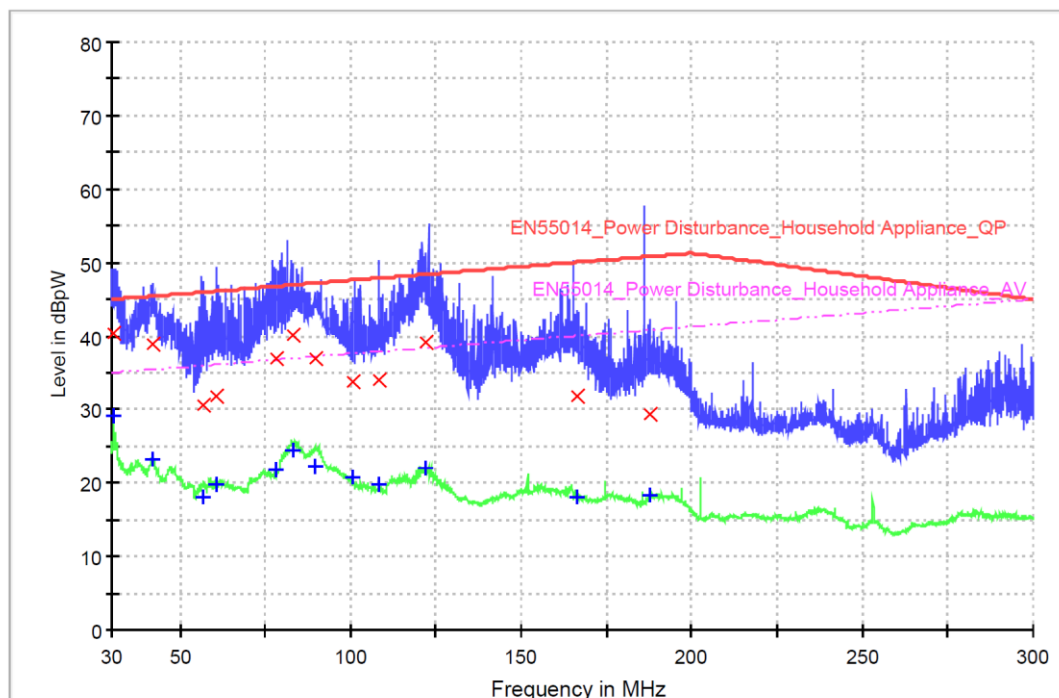
Frequency (MHz)	QuasiPeak (dBpW)	Meas. Time (ms)	Bandwidth (kHz)	Corr. (dB)	Margin - QPK (dB)	Limit - QPK (dBpW)
30.600000	36.6	1000.0	120.000	10.9	8.5	45.0
38.120000	33.2	1000.0	120.000	10.1	12.1	45.3
48.960000	31.5	1000.0	120.000	9.6	14.2	45.7
55.480000	29.8	1000.0	120.000	8.9	16.2	45.9
124.760000	32.5	1000.0	120.000	7.4	16.0	48.5
158.160000	29.2	1000.0	120.000	6.5	20.6	49.7
188.680000	30.0	1000.0	120.000	5.1	20.9	50.9
218.960000	27.1	1000.0	120.000	5.7	23.0	50.1

Limit and Margin-AV

Frequency (MHz)	Average (dBpW)	Meas. Time (ms)	Bandwidth (kHz)	Corr. (dB)	Margin - AVG (dB)	Limit - AVG (dBpW)
30.600000	27.6	1000.0	120.000	10.9	7.5	35.0
38.120000	20.3	1000.0	120.000	10.1	15.0	35.3
48.960000	17.7	1000.0	120.000	9.6	18.0	35.7
55.480000	18.4	1000.0	120.000	8.9	17.6	35.9
124.760000	20.9	1000.0	120.000	7.4	17.6	38.5

Frequency (MHz)	Average (dBpW)	Meas. Time (ms)	Bandwidth (kHz)	Corr. (dB)	Margin - AVG (dB)	Limit - AVG (dBpW)
158.160000	15.8	1000.0	120.000	6.5	23.9	39.7
188.680000	15.0	1000.0	120.000	5.1	25.8	40.9
218.960000	14.8	1000.0	120.000	5.7	27.2	42.0

Figure 6: Spectral Diagrams, Power Disturbance, AC Mains, 30–300MHz for LW-F6002A3



Limit and Margin-QP

Frequency (MHz)	QuasiPeak (dBpW)	Meas. Time (ms)	Bandwidth (kHz)	Corr. (dB)	Margin - QPK (dB)	Limit - QPK (dBpW)
30.600000	40.3	1000.0	120.000	10.9	4.8	45.0
42.080000	38.9	1000.0	120.000	9.9	6.5	45.4
56.880000	30.5	1000.0	120.000	9.1	15.5	46.0
60.720000	31.8	1000.0	120.000	9.3	14.4	46.1
78.160000	36.9	1000.0	120.000	9.1	9.9	46.8
83.040000	40.2	1000.0	120.000	8.4	6.8	47.0
89.440000	36.9	1000.0	120.000	8.1	10.3	47.2
100.600000	33.7	1000.0	120.000	7.7	13.9	47.6
108.160000	33.9	1000.0	120.000	7.1	14.0	47.9
121.760000	39.2	1000.0	120.000	7.5	9.2	48.4
166.320000	31.7	1000.0	120.000	5.7	18.3	50.1
187.480000	29.3	1000.0	120.000	5.2	21.5	50.8

Limit and Margin-AV

Frequency (MHz)	Average (dBpW)	Meas. Time (ms)	Bandwidth (kHz)	Corr. (dB)	Margin - AVG (dB)	Limit - AVG (dBpW)
30.600000	29.1	1000.0	120.000	10.9	5.9	35.0

Frequency (MHz)	Average (dBpW)	Meas. Time (ms)	Bandwidth (kHz)	Corr. (dB)	Margin - AVG (dB)	Limit - AVG (dBpW)
42.080000	23.3	1000.0	120.000	9.9	12.1	35.4
56.880000	18.2	1000.0	120.000	9.1	17.8	36.0
60.720000	19.9	1000.0	120.000	9.3	16.2	36.1
78.160000	21.9	1000.0	120.000	9.1	14.9	36.8
83.040000	24.4	1000.0	120.000	8.4	12.6	37.0
89.440000	22.3	1000.0	120.000	8.1	14.9	37.2
100.600000	20.8	1000.0	120.000	7.7	16.8	37.6
108.160000	19.9	1000.0	120.000	7.1	18.0	37.9
121.760000	22.1	1000.0	120.000	7.5	16.3	38.4
166.320000	18.2	1000.0	120.000	5.7	21.9	40.0
187.480000	18.4	1000.0	120.000	5.2	22.5	40.8

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5.2.2 Radiated Disturbance in the Frequency Range from 30MHz to 1000MHz

Result:	Pass
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Port : Enclosure
Basic Standard : EN IEC 55014-1:2021 and CISPR16-2-3
Frequency Range : 30 –1000MHz
Limit : EN IEC 55014-1:2021, clause 4.3.4.5, Table 9.

According to a) of clause 4.3.4.2 of EN IEC 55014-1:2021:

“The EUT shall be also deemed to comply with the requirement of this document in the frequency range from 300 MHz to 1000 MHz without further testing if both conditions 1) and 2) below are fulfilled:”

1): the disturbance power emission from the EUT is lower than the limits of Table 7 reduced by the values of Table 8;

2): the maximum clock frequency shall be less than 30MHz.

Because the EUT meets the two conditions mentioned above, the EUT is deemed to meet the radiated requirements without actual testing.

6 Test Results I M M U N I T Y

According to the electrical characteristics above and EN IEC 55014-2:2021, the EUT belongs to category II equipment.

“Category I: equipment containing no electronic control circuitry.”

“Category II: mains operated equipment containing electronic control circuitry with no clock frequency higher than 15 MHz.”

“Category III: battery operated equipment not included in Category I.”

“Category IV: mains operated equipment containing electronic control circuitry with a highest clock frequency greater than 15 MHz but lower than or equal to 200 MHz.”

“Category V: mains operated equipment containing electronic control circuitry with a highest clock frequency greater than 200 MHz.”

During the immunity tests, the EUT was operated under conditions specified by clause 3.1 of this report.

Performance criterion A: The apparatus shall continue to operate as intended during the test. No degradation of performance or loss of function is allowed below a performance level (or permissible loss of performance) specified by the manufacturer, when the apparatus is used as intended.

Performance criterion B: The apparatus shall continue to operate as intended after the test. No degradation of performance or loss of function is allowed below a performance level (or permissible loss of performance) specified by the manufacturer, when the apparatus is used as intended. During the test, degradation of performance is allowed.

Performance criterion C: Temporary loss of function is allowed, provided the function is self-recoverable or can be restored by the operation of the controls, or by any operation specified in the instructions for use.

The EMC immunity performances of the EUT were tested according to EN IEC 55014-2:2021.

Date of testing: 2026.05.14-2026.05.15

Room temperature	:	18-23°C
Relative humidity	:	45-50%
Atmospheric pressure	:	102.1 kPa

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6.1 Enclosure

6.1.1 Electrostatic Discharge

Result:	Pass
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The immunity against electrostatic discharge was tested in accordance with EN IEC 55014-2:2021. Test setup and ESD-Generator are according to EN 61000-4-2 which is specified by EN IEC 55014-2:2021.

The EUT is placed on 0,8m wood table above the ground plane. And the minimum distance between the EUT and all other conductive structures except the ground plane beneath the EUT is more than 0,5m.

The reference ground plane is an aluminium sheet of 0.25mm minimum thickness. The reference ground plane is connected to the protective earth. The size of the ground plane is 2m × 2m.

A horizontal coupling plane (HCP), 1.6m × 0.8m, is placed on the table and isolated from the EUT and cables by an insulating support 0.5mm thick. Vertical coupling plane (VCP) of dimensions 0.5m × 0.5m is placed parallel to and positioned at a distance of 0.1m from the EUT.

Charge voltage : ±4.0kV (Contact Discharge), ±8.0kV (Air Discharge)
Polarity : positive / negative
Number of discharges : ≥10
Performance criteria : B

Table 8: ESD, Positive / Negative Polarity

Position	Kind of Discharge	Result	Remarks
Nonmetal Enclosure	Air discharge ±8kV	Pass	No disturbance of function
Button	Air discharge ±8kV	Pass	No disturbance of function
Metal Enclosure	Contact discharge ±4kV	Pass	No disturbance of function
Coupling plane (Both HCP and VCP)	Contact discharge ±4kV	Pass	No disturbance of function

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6.2 Input and Output AC Power Ports

6.2.1 Fast Transients on AC Power Lines

Result:	Pass
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The immunity against fast transients on AC power lines was tested in accordance to EN IEC 55014-2:2021. Test setup and the fast transient noise generator are according to EN 61000-4-4 which is specified by EN IEC 55014-2:2021.

The EUT is placed on 0.1m wood table above the ground plane. And the minimum distance between the EUT and all other conductive structures except the ground reference plane is more than 0.5m.

The length between the coupling device and the EUT is less than 1m. The cord length more than 1m, the excess length of the cable shall gathered into a flat coil with a 0,4m diameter, and situated at a distance of 0,1m above the ground reference plane.

The reference ground plane is an aluminium sheet of 0.25mm minimum thickness. The reference ground plane is connected to the protective earth. The size of the ground plane is 2m × 2m.

Test Voltage	: 1kV
Polarity	: negative/positive
Repetition frequency	: 5kHz
Test duration	: ≥120sec
Tr/Tn	: 5ns/50ns
Performance criteria	: B

Table 9: Burst, AC Power lines, Positive and Negative Polarity, for model LW-F6002C2

Coupling Method: Direct Injection		
Coupling Port	Test Voltage / Result	Remark
AC mains: L1 (L), L2 (N), PE	±1000V Pass	No disturbance of function

Table 10: Burst, AC Power lines, Positive and Negative Polarity, for model LW-F6002A3

Coupling Method: Direct Injection		
Coupling Port	Test Voltage / Result	Remark
AC mains: L1 (L), L2 (N), --	±1000V Pass	No disturbance of function

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6.2.2 Injected Current into AC Power Port

Result:	Pass
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The immunity against injected current into AC power port was tested according to EN IEC 55014-2:2021 in a shielded room. The Test setup and the test generator are according to EN 61000-4-6 which is specified by EN IEC 55014-2:2021.

The EUT is placed on 0,1m wood table above the ground plane. And the minimum distance between the EUT and all other conductive structures except the reference ground plane is more than 0.5m.

The EUT comprised a single unit. The coupling and decoupling networks were inserted on the power supply connection. The coupling and decoupling networks was placed on the ground reference plane, making direct contact with it at about 0.1-0.3 meter from EUT. The cable between EUT and CDN is as short as possible and not bundled nor wrapped. The height of cable between the EUT and the coupling and decoupling networks above the ground reference plane was 50mm.

Voltage Level	: 3V(rms)(unmodulated)
Environmental phenomena	: r.f. current, common mode, 1kHz, 80%AM
Source impedance	: 150 Ω
Frequency range	: 0.15 - 230 MHz
Sweeping rate	: $\leq 1,5 \times 10^{-3}$ decades/s
Performance criteria	: A

Table 11: Injected current, AC Power Port, for model LW-F6002C2

Coupling Port	Coupling Method:	Result	Remark
AC mains: L1 (L), L2 (N), PE	CDN M-3	Pass	No disturbance of function

Table 12: Injected current, AC Power Port, for model LW-F6002A3

Coupling Port	Coupling Method:	Result	Remark
AC mains: L1 (L), L2 (N), --	CDN M-2	Pass	No disturbance of function

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6.2.3 Surges to AC Power Port

Result:	Pass
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The immunity against surges to AC power port was tested in accordance to EN IEC 55014-2:2021. Test setup and the Combination Wave Generator (CWG) are according to EN 61000-4-5 which is specified by EN IEC 55014-2:2021.

Open-circuit Test Voltage	: 1 kV (phase to neutral)/2 kV (Phase/neutral to PE)
Tr/Tn	: 1.2/50µs (open-circuit voltage) 8/20µs (short-circuit current)
Test numbers	: 5 positive and 5 negative pulses
Test angle	: 90° and 270°
Repetition rate	: 1 surge/min
Performance criteria	: B

Table 13: Surges to AC Power lines, positive/negative, for model LW-F6002C2

Line	Tested voltage/coupling phase	Result	Remarks
Phase to neutral	+1.0kV, $+\pi/2$ (5 times)	Pass	No disturbance of function
	-1.0kV, $-\pi/2$ (5 times)	Pass	No disturbance of function
Phase/neutral to PE	+2.0kV, $+\pi/2$ (5 times)	Pass	No disturbance of function
	-2.0kV, $-\pi/2$ (5 times)	Pass	No disturbance of function

Table 14: Surges to AC Power lines, positive/negative, for model LW-F6002A3

Line	Tested voltage/coupling phase	Result	Remarks
Phase to neutral	+1.0kV, $+\pi/2$ (5 times)	Pass	No disturbance of function
	-1.0kV, $-\pi/2$ (5 times)	Pass	No disturbance of function

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6.2.4 Voltage dips and interruptions to AC Power Port

Result:	Pass
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The immunity against voltage dips and interruptions to AC power port was tested in accordance to EN IEC 55014-2:2021. Test setup and the test generator are according to EN 61000-4-11 which is specified by EN IEC 55014-2:2021.

Table 15: Test condition and Test Result for Voltage dips and Short interruptions

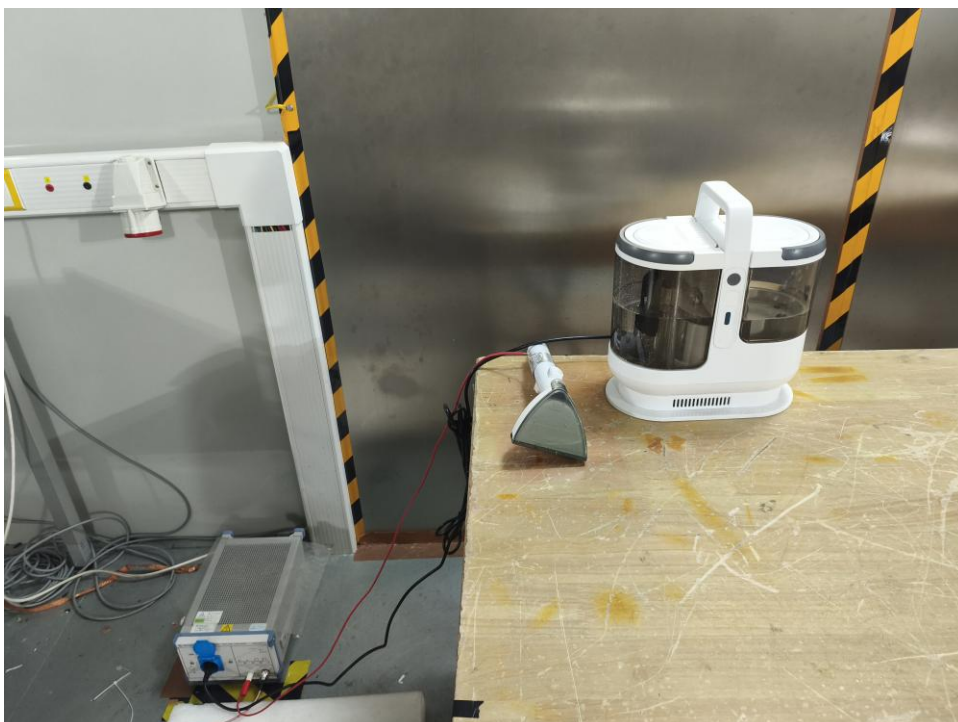
Test level (in % UT)	Duration	Performance criteria	Result	Remarks
0	0.5/0.5	C	Pass	During the test, the rotational speed decreased and returned to its initial state after the test was completed
40	10/12	C	Pass	During the test, the rotational speed decreased and returned to its initial state after the test was completed
70	25/30	C	Pass	During the test, the rotational speed decreased and returned to its initial state after the test was completed

7 Photographs of the Test Set-Up

Photograph 1: Set-up for Harmonics, Voltage Fluctuation and Flicker



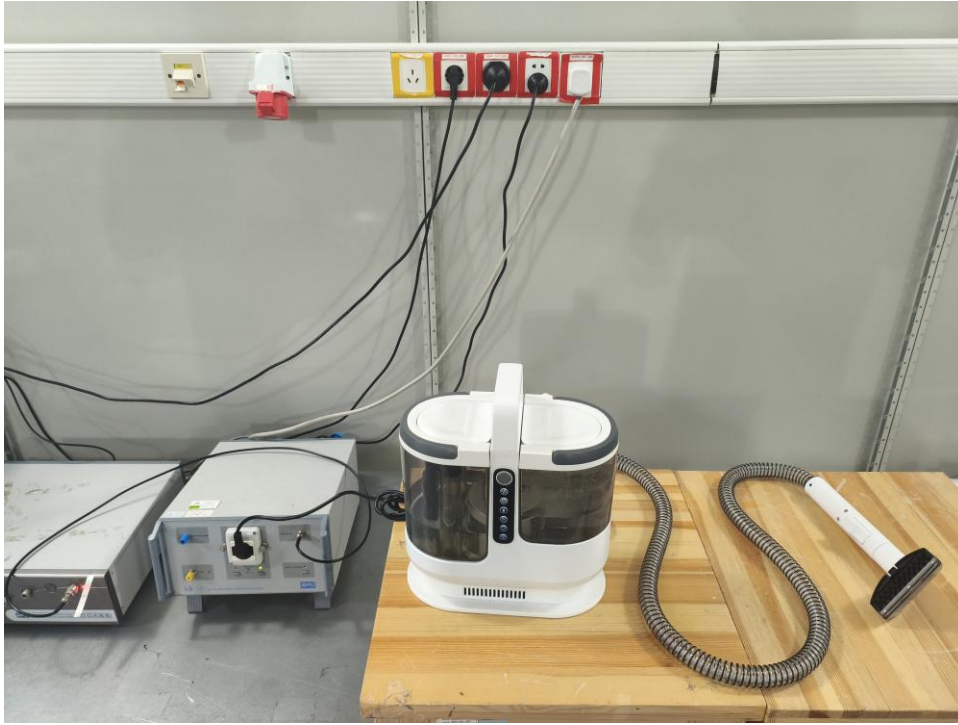
Photograph 2: Set-up for Disturbance Voltage



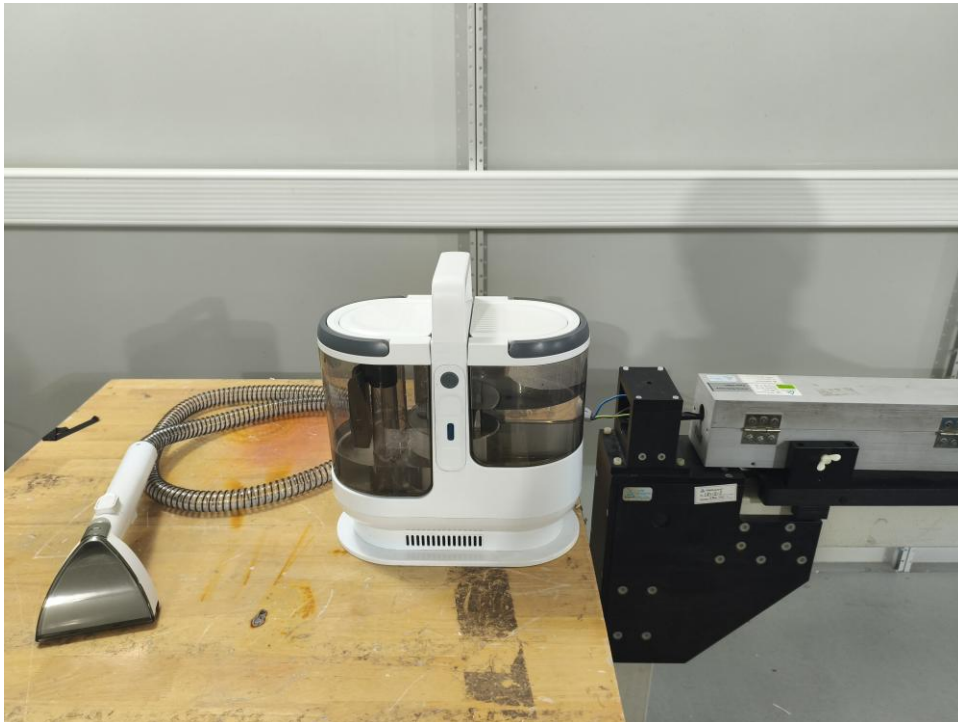
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Photograph 3: Set-up for Click



Photograph 4: Set-up for Disturbance Power



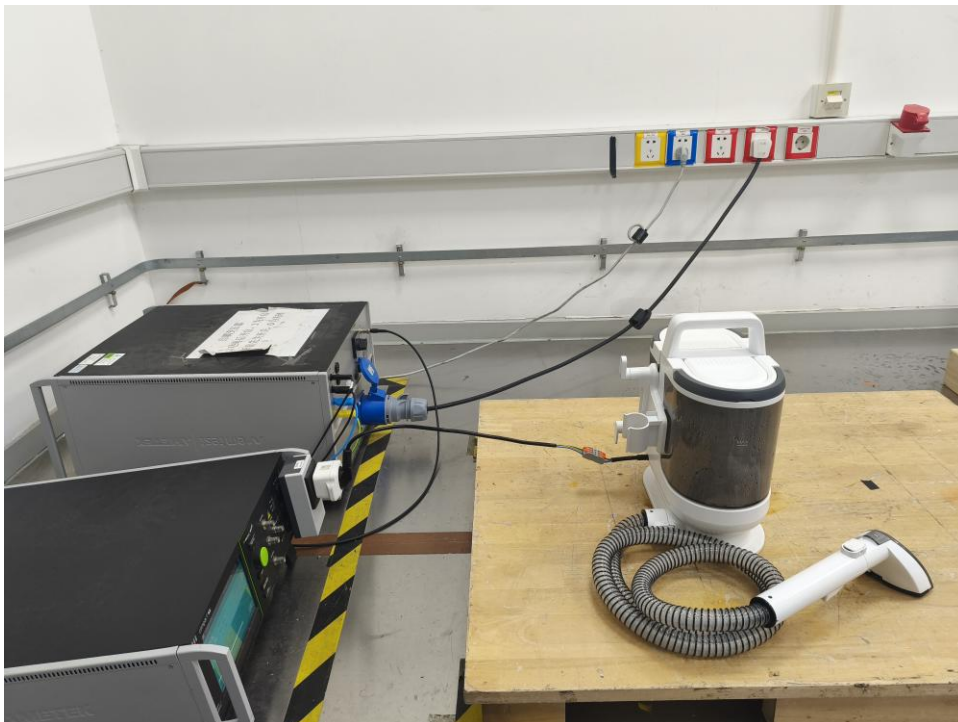
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Photograph 5: Set-up for ESD



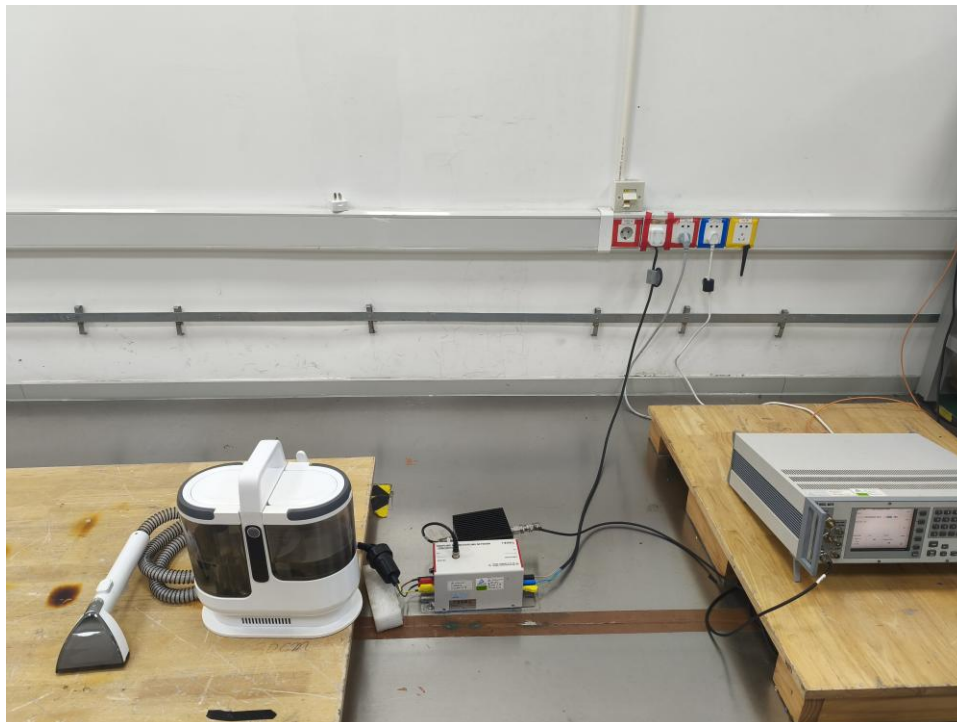
Photograph 6: Set-up for Surge, EFT and Voltage Dips



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Photograph 7: Set-up for Injected Current



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